

LLM Integration: Autonomous Causal Estimation Pipeline

This directory contains the integration of Large Language Models (LLMs) into the AD_regulome heritability analysis pipeline. The objective is to utilize the parametric memory of advanced AI models to generate literature-grounded hypotheses for significant cell types across diverse genomic datasets.

1. Model Selection & Comparative Analysis

During the development phase, we performed a head-to-head comparison between two primary candidates:

- Gemma4-Biology Specialist (Local): A 14B parameter model surgically pruned for biological reasoning.
- Gemini 2.5 Flash (Cloud): A high-performance generalist model.

The Conclusion: While the local specialist demonstrated strong domain knowledge, Gemini 3.5 Flash was selected as the primary engine for this project. Gemini provided superior taxonomic precision, correctly identifying specific anatomical sub-compartments (e.g., Atrial vs. Ventricular) where the local model provided more generalized results. Furthermore, Gemini demonstrated better logical consistency and structural reliability in generating machine-readable JSON outputs.

2. Universal Prompt Architecture

We developed a "Senior Scientist" prompt designed to be organ-agnostic and compatible with any phenotype in the GWAS Catalog.

3. Case Study: Atrial Fibrillation (PMID: 30061737)

To verify the system, we compared the LLM estimation against the published heritability results in the Zhang et al. paper.

Source	Rank 1	Rank 2	Rank 3
Published Paper	Adult A Cardiomyocyte	Adult V Cardiomyocyte	Fetal A Cardiomyocyte
LLM Estimation	Adult A Cardiomyocyte	Fetal A Cardiomyocyte	Adult V Cardiomyocyte

Analysis of the Order Difference: The LLM correctly identified the top three cell types but prioritized Fetal A Cardiomyocytes over the Adult V variant. This occurs because the LLM’s

"Knowledge Base" is built on the global consensus of scientific literature. In current research, the link between atrial fibrillation and developmental electrophysiological programming (Fetal stage) is a highly discussed and "high-signal" topic. The LLM reflects this literature density, prioritizing the developmental drivers of the heart's conduction system.

4. Execution Steps

To run the autonomous estimation, follow this sequence within the notebook:

1. Environment Setup: Run the initialization cell to load the ollama, google.genai, and json libraries, and configure the secure .env API keys.
2. Configuration: Update the GWAS_FILE text box with your target phenotype and PMID (e.g., Type 2 diabetes (30297969)).
3. Prompt Generation: Execute the prompt cell to build the study-specific instructions.
4. Model Inference: Run the Model Execution cell to retrieve the structured biological hypothesis.